JavaScript

```javascript
function Counter() {
  const [count, setCount] = useState(0);

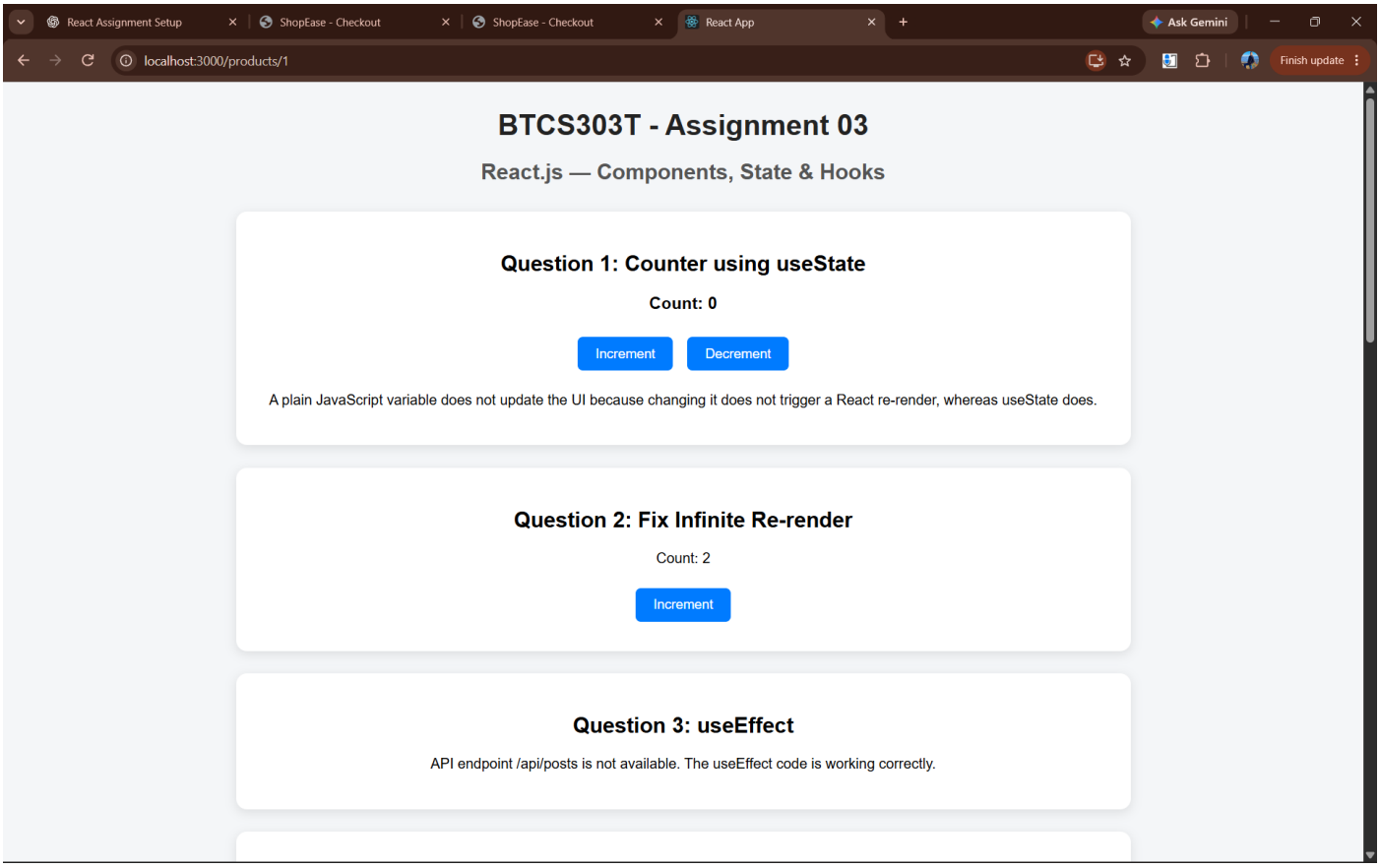
  return (
    <div className="question">
      <h2>Question 1: Counter using useState</h2>

      <h3>Count: {count}</h3>

      <button onClick={() => setCount(count + 1)}>
        Increment
      </button>

      <button onClick={() => setCount(count - 1)}>
        Decrement
      </button>

      <p>
        A plain JavaScript variable does not update the UI because changing it
        does not trigger a React re-render, whereas useState does.
      </p>
```



JavaScript

```
function CounterFix() {
  const [count, setCount] = useState(0);

  return (
    <div className="question">
      <h2>Question 2: Fix Infinite Re-render</h2>

      <p>Count: {count}</p>

      <button onClick={() => setCount(count + 1)}>
        Increment
      </button>
    </div>
  );
}
```

Question 2: Fix Infinite Re-render

Count: 2

Increment

```
function Posts() {
  const [posts, setPosts] = useState([]);

  useEffect(() => {
    fetch("/api/posts")
      .then((response) => response.json())
      .then((data) => {
        setPosts(data);
      })
      .catch((error) => {
        console.error("Error fetching posts:", error);
      });
  }, []);

  return (
    <div>
      <h2>Question 3: useEffect</h2>

      {posts.map((post) => (
        <div key={post.id}>
          <h3>{post.title}</h3>
```

Question 3: useEffect

API endpoint /api/posts is not available. The useEffect code is working correctly.

JavaScript

```
function PriceInput({ price, setPrice }) {
  return (
    <div>
      <h3>Enter Price</h3>

      <input
        type="number"
        value={price}
        onChange={(e) => setPrice(e.target.value)}
      />
    </div>
  );
}

function PriceDisplay({ price }) {
  return <p>Price: ₹{price}</p>;
}

function PriceApp() {
  const [price, setPrice] = useState(0);

  return (
```

Question 4: Lifting State Up

Enter Price

64000

Price: ₹64000

JavaScript

```javascript
function useLocalStorage(key, initialValue) {
  const [value, setValue] = useState(() => {
    const storedValue = localStorage.getItem(key);

    if (storedValue !== null) {
      return JSON.parse(storedValue);
    }

    return initialValue;
  });

  const setStoredValue = (newValue) => {
    setValue(newValue);
    localStorage.setItem(key, JSON.stringify(newValue));
  };

  return [value, setStoredValue];
}
```

Question 5: useLocalStorage

AYUSH SANWAL

Stored Name: AYUSH SANWAL

JavaScript

```
function ItemList() {
  const items = [
    { id: 1, name: "Laptop" },
    { id: 2, name: "Mobile Phone" },
    { id: 3, name: "Keyboard" },
    { id: 4, name: "Mouse" }
  ];

  return (
    <div className="question">
      <h2>Question 6: List Keys</h2>

      <ul>
        {items.map((item) => (
          <li key={item.id}>
            {item.name}
          </li>
        ))}
      </ul>
    </div>
  );
}
```

Question 6: List Keys

- Laptop
- Mobile Phone
- Keyboard
- Mouse

Button.js

JavaScript

```
function Button({ variant, children }) {
  return (
    <button className={`btn ${variant}`}>
      {children}
    </button>
  );
}
```

ButtonDemo.js

JavaScript

```javascript
function ButtonDemo() {
  return (
    <div className="question">
      <h2>Question 7: Reusable Button</h2>

      <Button variant="primary">
        Save
      </Button>

      <Button variant="danger">
        Delete
      </Button>
    </div>
  );
}
```



Question 7: Reusable Button

Save Delete

JavaScript

```
    </div>
  );
});

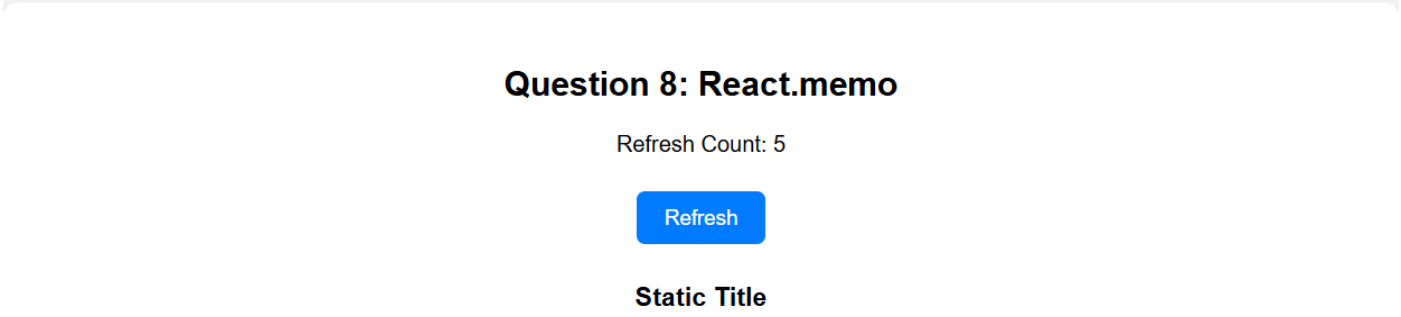
function ParentDashboard() {
  const [tick, setTick] = useState(0);

  return (
    <div className="question">
      <h2>Question 8: React.memo</h2>

      <p>Refresh Count: {tick}</p>

      <button onClick={() => setTick(tick + 1)}>
        Refresh
      </button>

      <ChildCard title="Static Title" />
    </div>
  );
}
```



JavaScript

```
<BrowserRouter>

  <nav className="navbar">
    <Link to="/">Home</Link>
    <Link to="/products">Products</Link>
    <Link to="/products/1">Product 1</Link>
  </nav>

  <Routes>

    <Route
      path="/"
      element={<Home />}
    />

    <Route
      path="/products"
      element={<Products />}
    />

    <Route
```

Question 9: React Router v6

Home Products Product 1

Product Details

Product ID: 1

JavaScript

```javascript
const [formData, setFormData] = useState({
  name: "",
  email: "",
  phone: ""
});

const handleChange = (e) => {
  const { name, value } = e.target;

  setFormData({
    ...formData,
    [name]: value
  });
};
```

JavaScript

```javascript
<form onSubmit={handleSubmit}>

 <input
   type="text"
   name="name"
   value={formData.name}
   onChange={handleChange}
   placeholder="Enter name"
 />

 <input
   type="email"
   name="email"
   value={formData.email}
   onChange={handleChange}
   placeholder="Enter email"
 />

 <input
   type="tel"
   name="phone"
   value={formData.phone}
```

Question 10: Form State

AYUSH SANWAL

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Submit